UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE O Level

MARK SCHEME for the November 2005 question paper

5090 BIOLOGY

5090/03 Paper 3 maximum raw mark 40

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1 (a) Drawing marks: **[D.3]**

1 Clear, clean realistic drawing, 5 cm or more, with basal disc.
2 Fleshy scale leaves accurately shown and attached.
3 Papery covering scales shown

Labels: Any 2 of: leaf/**scale**/bud/stem/disc; **[for 1]**

(b) (i) Up to three from:

both are roots;
transverse sections;
central conducting tissue;
surrounded by storage tissue;
AVP; **[up to 3]**

(ii) Expect a comparison/contrast to be made **[1 mark per line]**

specimen in Fig.1.1	cut surface of W2;
cellular structure clear	no cells visible;
vegetative organ AW	storage tissue/organ;
cells almost empty	full of starch/stored food; **[up to 2]**

(c) (i) black due to starch;
central pattern;
detail of distribution;
ref. time, or development of reaction; **[up to 3]**

(ii) Test: Heat/boil with Benedict's; how heated;
obs: orange/reddish (ppt);
A: green if ref dilute concn.
concl: ref. (amount of) R.S/glucose present; **[4]**

(iii) cutting open cells;
increased surface area;
releasing contents/sugar;
decanting gets rid of (masking) colour AW; **[up to 2]**

[Total:18]

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2 (a) (i) Drawing marks; **[D.2]**

1 **Ruled** rectangle and square, in proportion.
2 Specimen well placed.

Labels: slide, cover glass, specimen, water; **[2]**

[3 correct = 2 marks 2 correct = 1 mark]

(ii) Two from:

air bubbles avoided;
central placement;
no overflow of water; **[2]**

(b) Filter paper soaks up water;
stain penetrates/drawn under cover;
stain emerges on f paper/replaces water;
specimen becomes stained;
ref. development with time; **[up to 4]**

(c) (i) Drawing marks: **[D.3]**

1 At least 6 cm clear and clean.
2 Large and small/simple leaves observed.
3 Leaf arrangement accurately shown.

Labels: stem; leaves; **[2]**

(ii) Line ruled and accurately measured with units once;
expression line length over specimen size;
correctly calculated and recorded; **[3]**

(iii) Four from:

specimen supported;
for examination with microscope;
clear for viewing/no fingers in the way AW;
specimen flat/thin;
ref. ease of focusing;
AVP; [such as ref. penetration of light/ref. visibility of staining] **[4]**

[Total: 22]

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